DEPARTMENT OF MATHEMATICS

MAT111 INTRODUCTORY MATHEMATICS I

WEEK 9: LECTURE NOTE

EXPONENTIAL AND LOGARITHMIC FUNCTIONS

EXPONENTIAL FUNCTIONS

Definition 1. *For $a > 0$, $a \neq 1$, the exponential function with base a is given by*

$$f(x) = a^x, \ x \in \mathbb{R}.$$

Example 2. *Evaluate $f(x) = 2^x$ at*
(a) $x = -3$ (b) $x = 0$ (c) $x = 0.5$ (d) $x = -\pi$
***Solution**: Use a calculator to get decimals rounded to 3 or 4 digits.*

(a) $f(-3) = 2^{-3} = \dfrac{1}{2^3} = \dfrac{1}{8} = 0.125$

(b) $f(0) = 2^0 = 1$

(c) $f(0.5) = 2^{0.5} = 2^{\frac{1}{2}} = \sqrt{2} = 1.4142$

(d) $f(-\pi) = 2^{-\pi} = \dfrac{1}{2^\pi} = 0.1133$

Example 3. *Evaluate $f(x) = (\frac{1}{2})^x$ at*
(a) $x = -3$ (b) $x = 0$ (c) $x = 0.5$ (d) $x = -\pi$
***Solution**:*

(a) $f(-3) = (\frac{1}{2})^{-3} = 8$

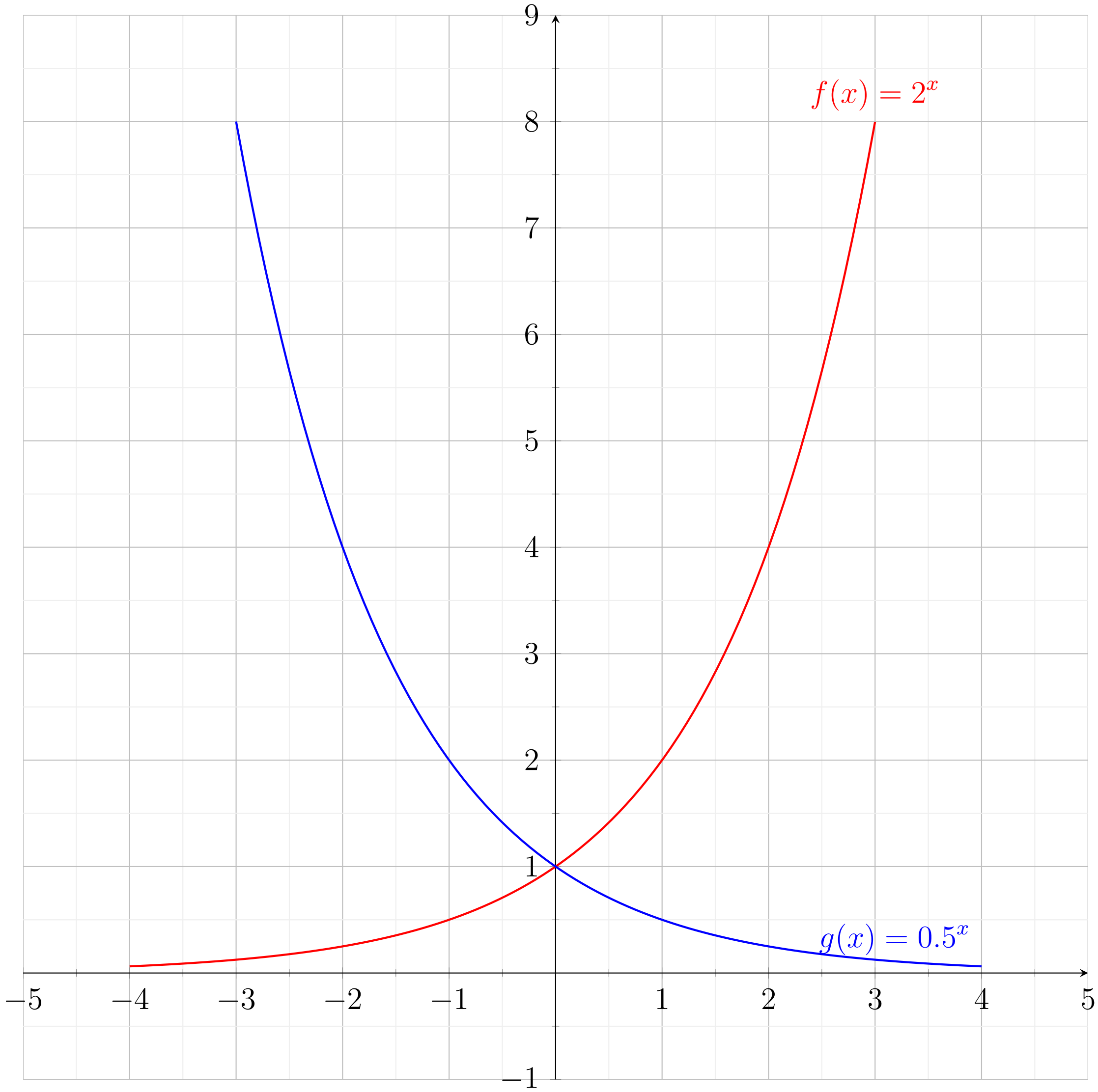
(b) $f(0) = (\frac{1}{2})^0 = 1$

(c) $f(0.5) = (\frac{1}{2})^{0.5} = \sqrt{\frac{1}{2}} = 0.7071$ *(use calculator)*

(d) $f(-\pi) = (\frac{1}{2})^{-\pi} = 8.8261$*(use calculator)*

Thoroughly look at the values of $f(x) = 2^x$ and $g(x) = (\frac{1}{2})^x$ to the nature of their graphs.

x	-5	-4	-3	-2	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1	2	3	4	5
f(x)	$\frac{1}{32}$	$\frac{1}{16}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	1	$\sqrt{2}$	2	4	8	16	32
g(x)	32	16	8	4	2	$\sqrt{2}$	1	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$



Domain f=$(-\infty, +\infty)$=Domain g

Range f =$(0, \infty)$=Range g

From the above graphs we have the following properties:

$f(x) = a^x, \quad a > 1$	$f(x) = a^x, \quad 0 < a < 1$
Domain =$(-\infty, +\infty)$ Range =$(0, +\infty)$ y-intercept: $(0, 1)$ Increasing on $(-\infty, +\infty)$ x-axis is a horizontal asymptote	Domain =$(-\infty, +\infty)$ Range =$(0, +\infty)$ y-intercept: $(0, 1)$ Decreasing on $(-\infty, +\infty)$ x-axis is a horizontal asymptote

Recall the following **properties of Exponents**:

(a) $a^x a^y = a^{x+y}$.

(b) $\dfrac{a^x}{a^y} = a^{x-y}$.

(c) $a^{-x} = \dfrac{1}{a^x} = (\dfrac{1}{a})^x$.

(d) $(ab)^x = a^x b^x$.

(e) $(a^x)^y = a^{xy}$.

(f) $(\dfrac{a}{b})^x = \dfrac{a^x}{b^x}$.

(g) $|a^2| = |a|^2 = a^2$.

(h) $a^0 = 1$.

TRANSFORMATIONS OF THE PARENT FUNCTION $f(x) = a^x$

Given the graph of $f(x) = a^x$ then

(i) $h(x) = f(-x) = a^{-x}$: Reflection of $f(x)$ about $y-$axis

(ii) $h(x) = -f(x) = -a^x$: Reflection of $f(x)$ about $x-$axis

(iii) $h(x) = f(x+c) = a^{x+c}$ $(c > 0)$: Horizontal shift of $f(x)$ c units **left**

(iv) $h(x) = f(x-c) = a^{x-c}$ $(c > 0)$: Horizontal shift of $f(x)$ c units **right**

(v) $h(x) = f(x) + c = a^x + c$ $(c > 0)$: Vertical shift of $f(x)$ c units **upwards**

(vi) $h(x) = f(x) - c = a^x - c$ $(c > 0)$: Vertical shift of $f(x)$ c units **downwards**

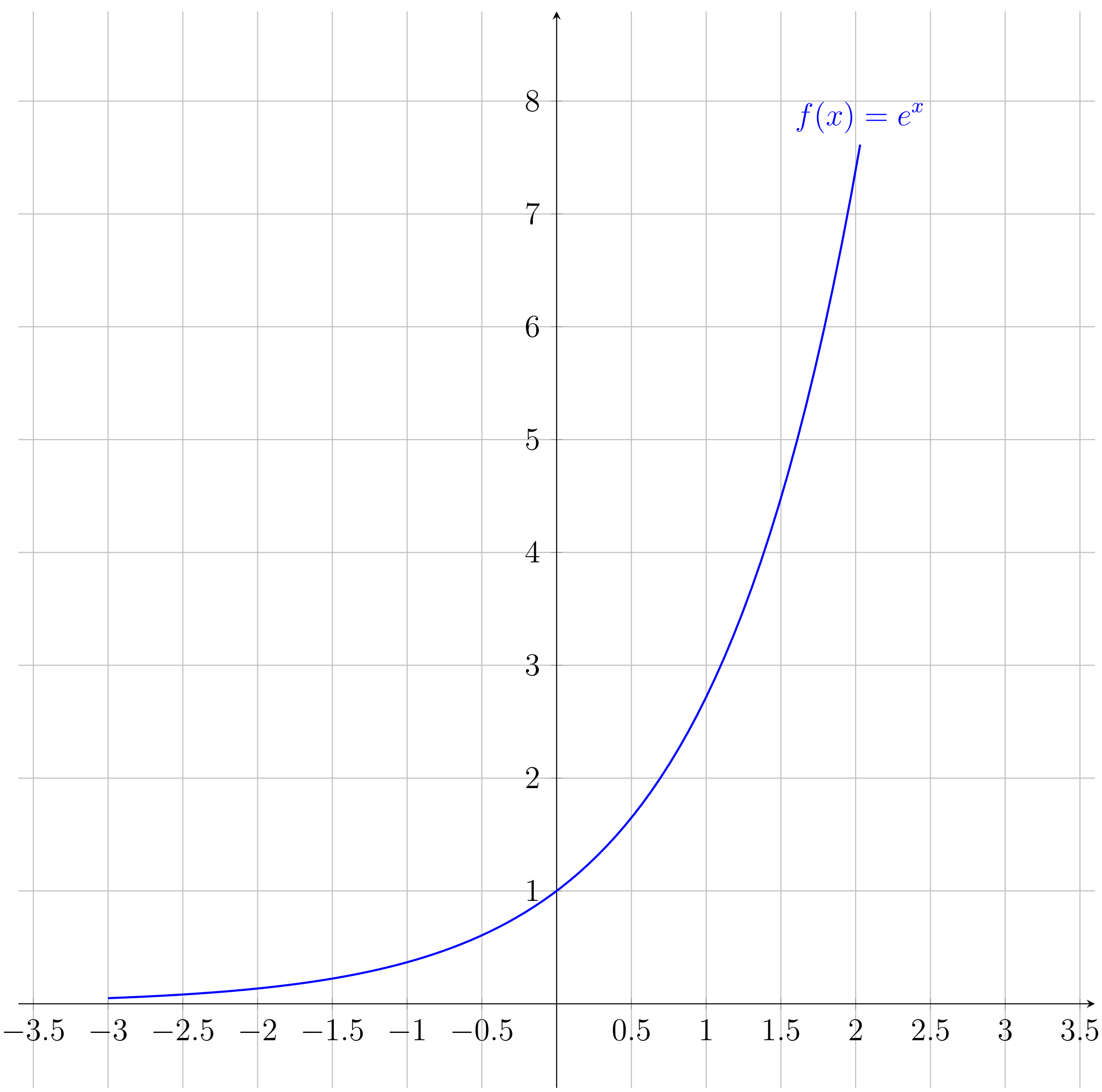
THE NATURAL BASE $e = 2.718281828...$

The irrational number $e = 2.718281828...$ is called **natural base** and the function

$$f(x) = e^x$$

is called the **natural exponential function**.

x	-3	-2	-1	-0.5	0	0.5	1	2
e^x	0.0498	0.1353	0.3679	0.6065	1	1.6487	e	7.3891



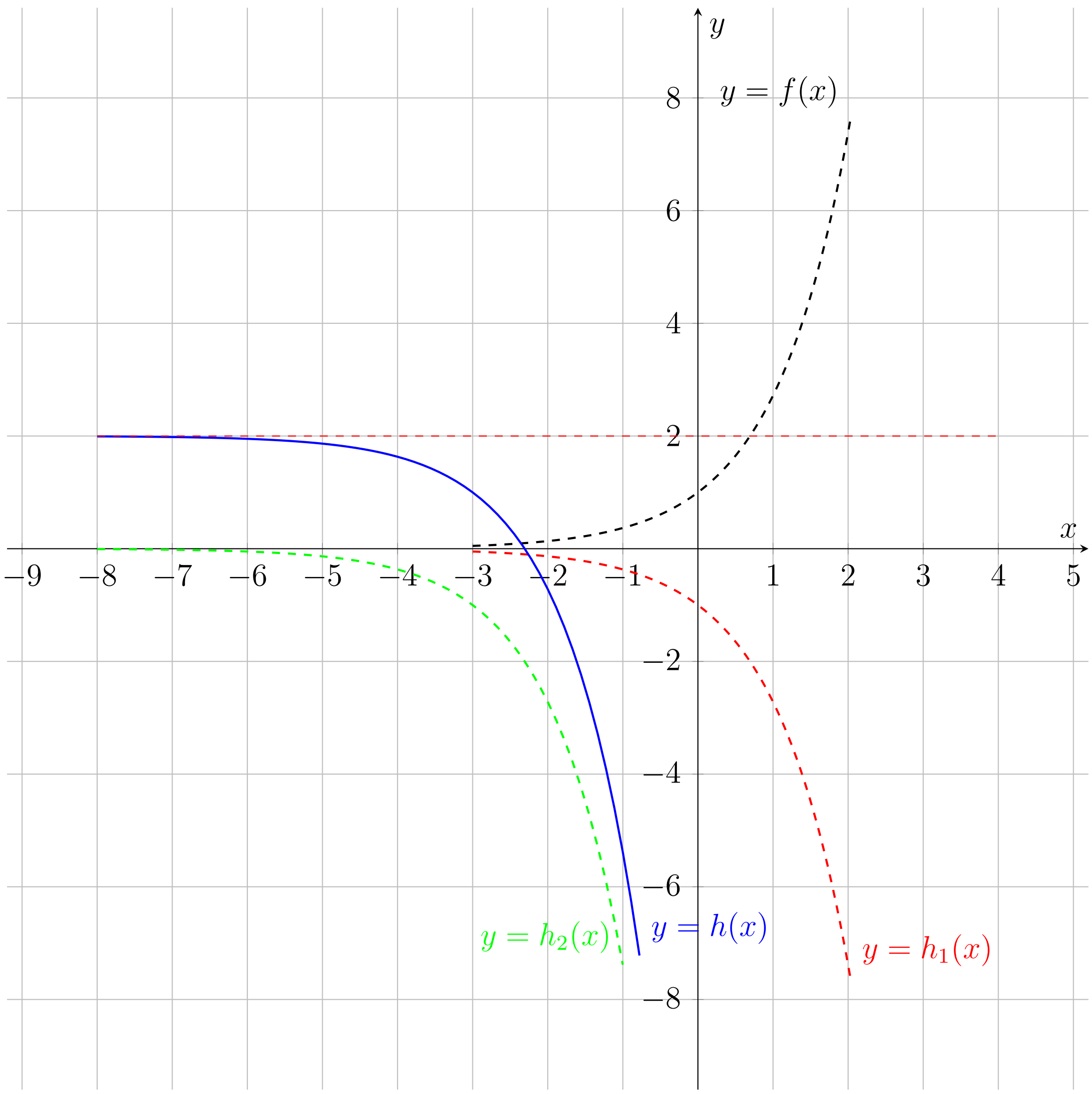
Example 4. *Given the function $h(x) = 2 - e^{x+3}$, describe all transformations in the graph $h(x)$ and state the parent function $f(x)$. Then sketch the graph of $h(x)$ and $f(x)$ on the same plane.*
Solution. *The parent function $f(x) = e^x$ is*

(i) *Reflected about $x-$axis: $h_1(x) = -f(x) = -e^x$*

(ii) *Horizontally shifted 3 units to the left: $h_2(x) = h_1(x+3) = -e^{x+3}$*

(iii) *Vertically shifted 2 units upwards: $h(x) = 2 + h_2(x) = 2 - e^{x+3}$*

GRAPHS OF $f(x) = e^x,\ h_1(x) = -e^x,\ h_2(x) = -e^{x+3},\ h(x) = 2 - e^{x+3}$



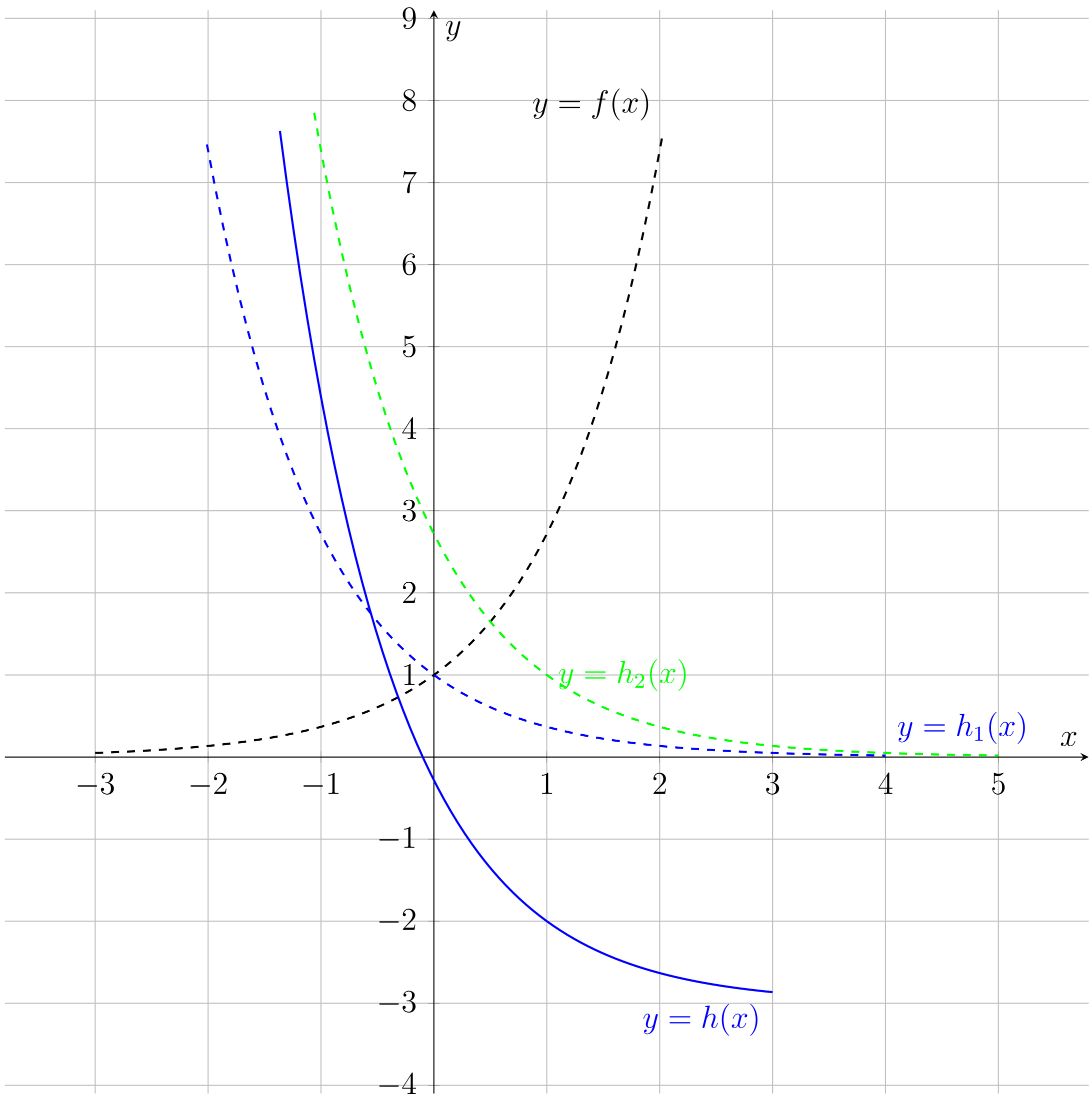
Example 5. *Given the function $h(x) = 2^{-x+1} - 3$, state the parent function $f(x)$ and describe all sequence of transformations to obtain $h(x)$. Then sketch the graph of $h(x)$ and $f(x)$ on the same plane.*
Solution. *The parent function $f(x) = 2^x$ is*

(i) *Reflected about $y-$axis: $h_1(x) = f(-x) = e^{-x}$*

(ii) *Horizontally shifted 1 unit to the right: $h_2(x) = h_1(x-1) = -e^{-(x-1)}$*

(iii) Vertically shifted 3 units downwards: $h(x) = h_2(x) - 3 = e^{-x+1} - 3$

(iv) ***GRAPHS OF*** $f(x) = 2^x, \;\; h_1(x) = 2^{-x}, \;\; h_2(x) = -e^{-(x-1)}, \;\; h(x) = e^{-x+1} - 3$



Exercises

1. Use a calculator to the given exponential function at the indicated value of x. Round your answer to three decimal places

   $(a)\;\; f(x) = 5^x, \;\; x = -\pi \qquad (b)\;\; f(x) = e^x, \;\; x = 9.2 \qquad (c)\;\; f(x) = \left(\frac{3}{4}\right)^x, \;\; x = -3$

2. Use the graph of the parent function $f(x)$ to describe all transformations that yield the graph of $g(x)$. Then sketch the graphs of $f(x)$ and $g(x)$ on the same plane.

   (a) $f(x) = 3^x, \;\; g(x) = 3^{x-5}$

   (b) $f(x) = 0.3^x, \;\; g(x) = -0.3^x + 5$

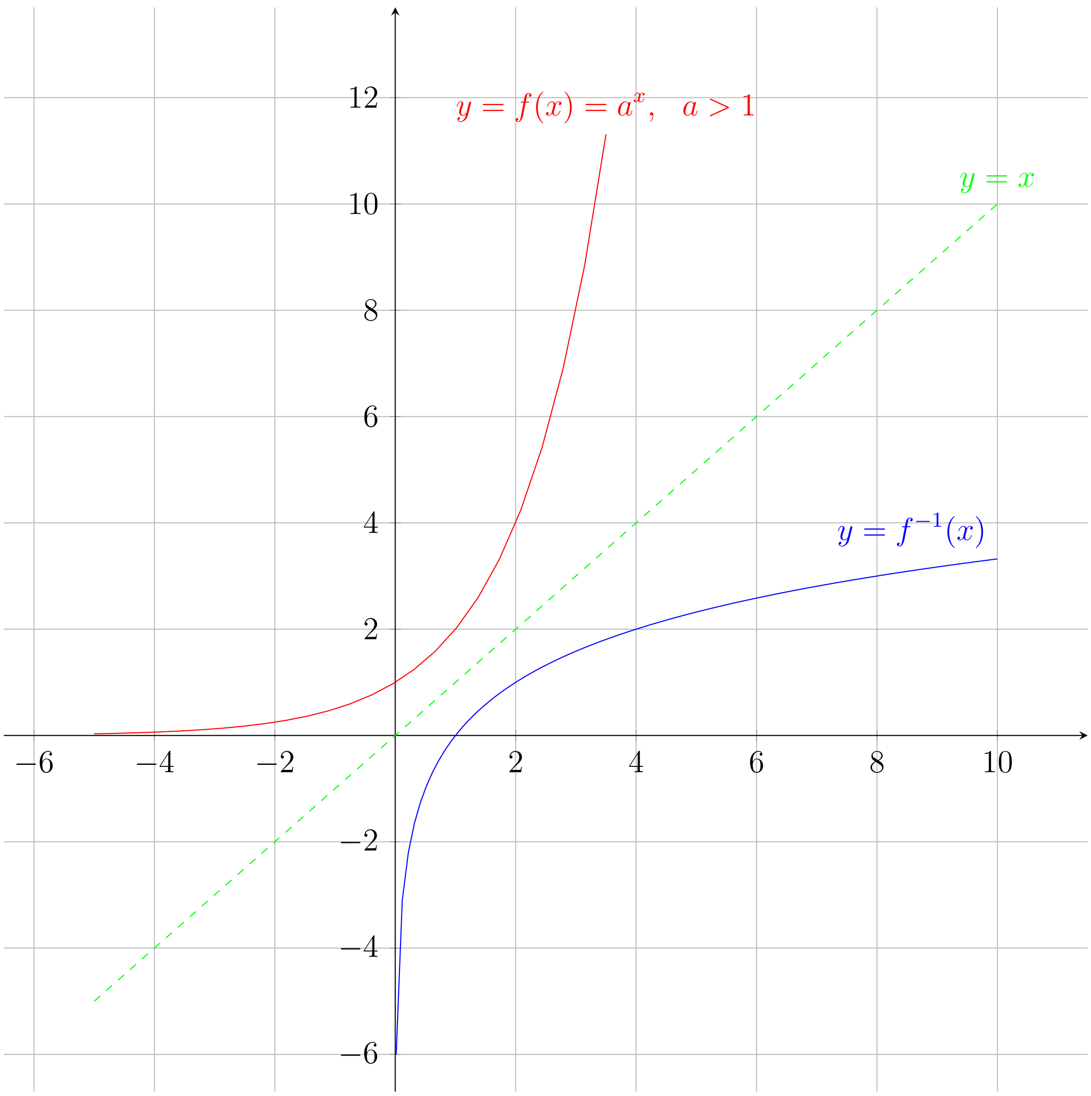
   (c) $f(x) = \left(\frac{1}{2}\right)^x, \;\; g(x) = \left(\frac{1}{2}\right)^{-(x+4)}$

LOGARITHMIC FUNCTIONS

We see that the graphs of the exponential function

$$f(x) = a^x, \quad a > 0, \quad a \neq 1$$

passes the **Horizontal Line Test** and therefore must have an inverse function as shown in the graph below.



Definition 6. *For $x > 0, \quad a > 0$ and $a \neq 1$, the inverse of exponential function $y = a^x$ is called the* ***logarithmic function with base*** *a, denoted by $y = \log_a x$. That is,*

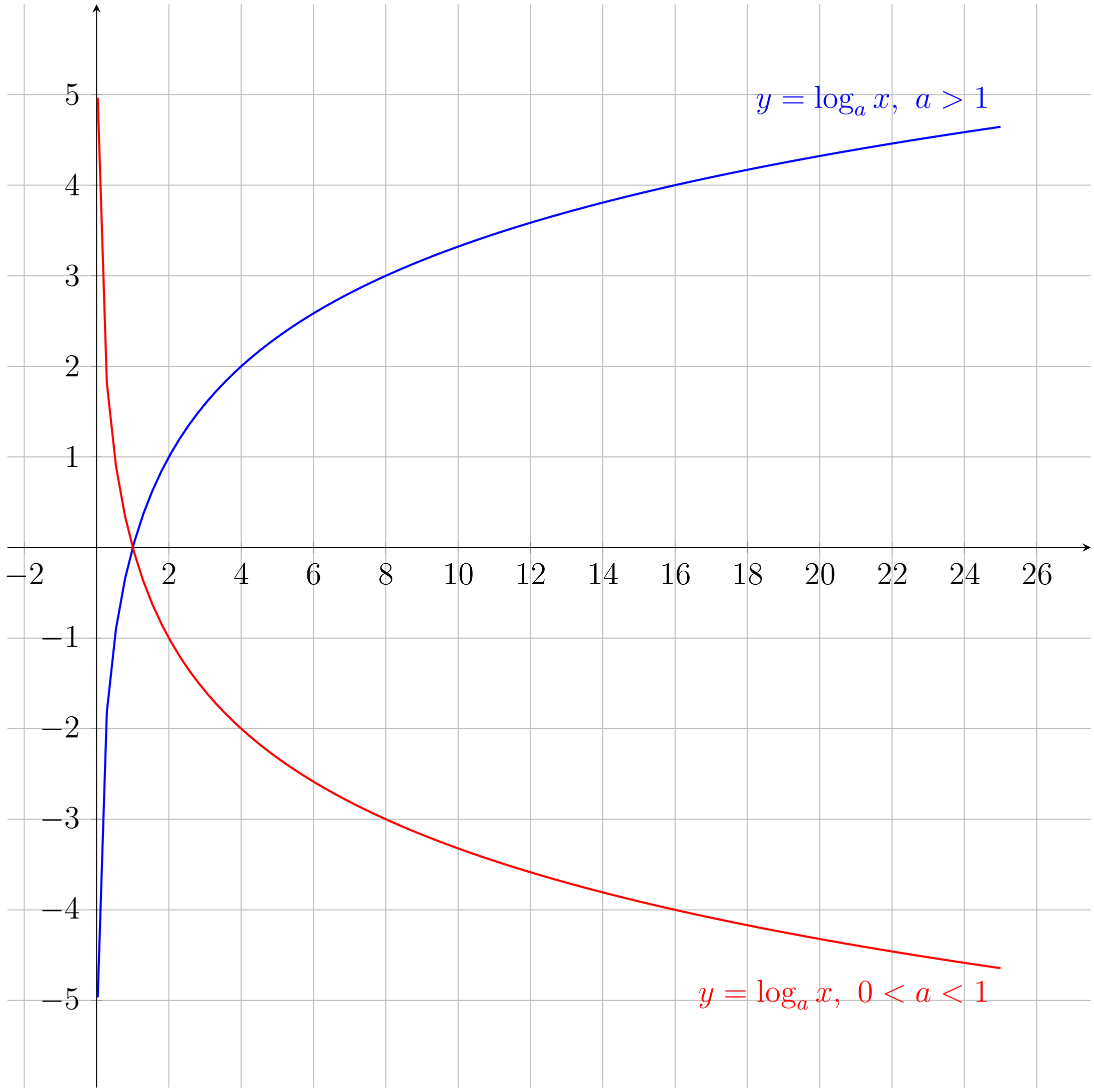
$$y = \log_a x \text{ if and only if } x = a^y.$$

Note that from the definition of inverse function, if $f(x) = a^x$ and $g(x) = \log_a x$, then we have:

1. *Domain of g = Range of f = $(0, +\infty)$.*

2. *Range of g =Domain of f =$(-\infty, +\infty)$.*

GRAPHS OF $f(x) = \log_2 x$ and $g(x) = \log_{\frac{1}{2}} x$



Example 7. *Evaluate $f(x) = \log_3 x$ at the given points in the table.*

x	0.001	0.1	0.5	1	1.5	2	3	4	9	75
$f(x)$	-6.288	-2.095	-0.631	0	0.369	0.631	1	1.261	2	3.930

Definition 8. *The inverse of the natural exponential function,*

$$y = e^x,$$

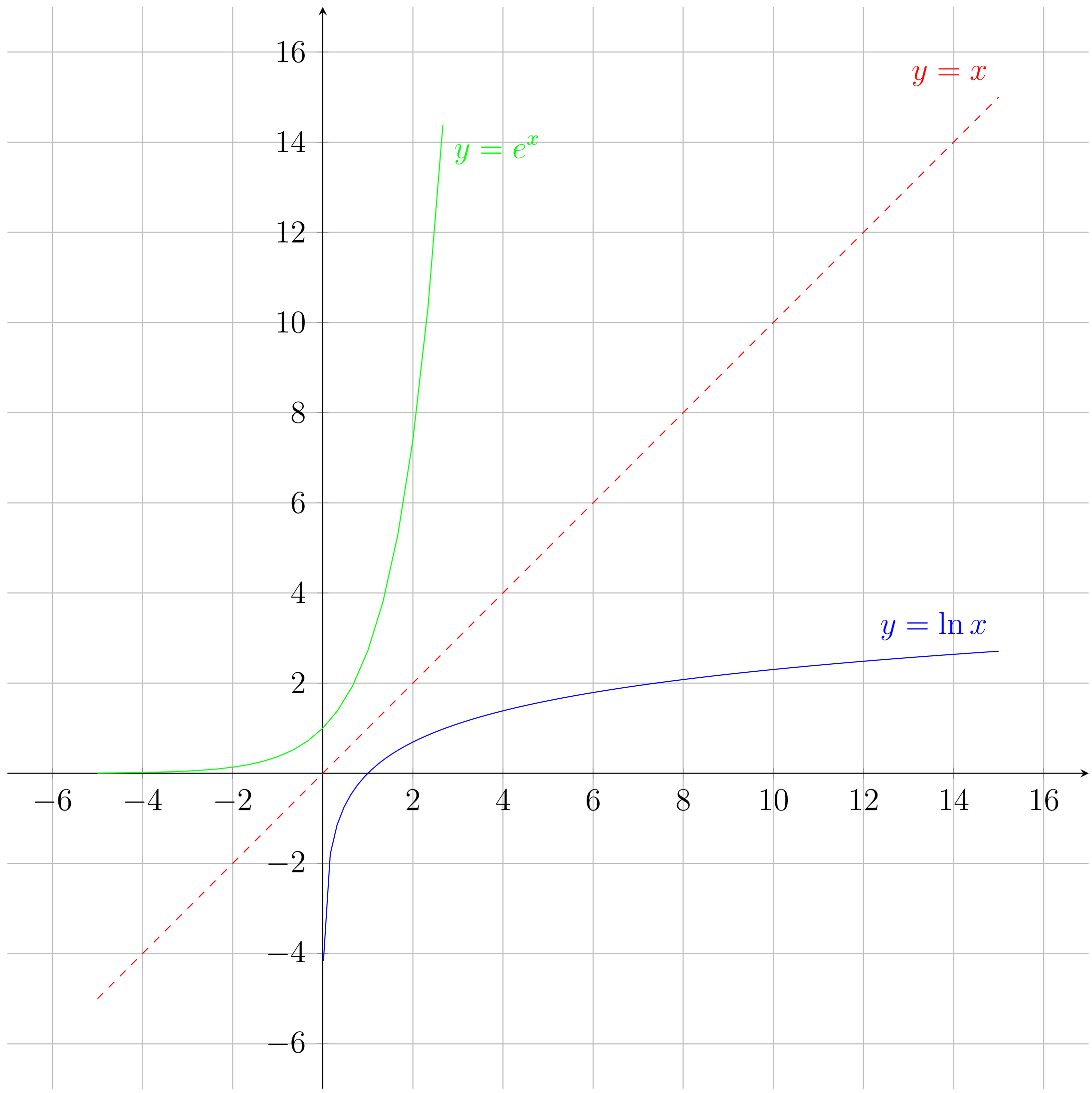
*is called the **NATURAL LOGARITHMIC** function. The natural logarithmic function is denoted by*

$$y = \ln x;$$

that is,

$$y = \ln x \text{ if and only if } x = e^y.$$

The graph of natural logarithmic function is given below:



Summary: Logarithmic functions have the following properties:

$f(x) = \log_a x, \quad a > 1$	$f(x) = \log_a x, \quad 0 < a < 1$
♦ Domain $=(0, +\infty)$ ♦ Range $=(-\infty, +\infty)$ ♦ x-intercept: $(1, 0)$ ♦ Increasing on $(0, +\infty)$ ♦ y-axis is a vertical asymptote ♦ It is one-to-one function	Domain $=(0, +\infty)$ Range $=(-\infty, +\infty)$ x-intercept: $(1, 0)$ Decreasing on $(0, +\infty)$ y-axis is a vertical asymptote It is one-to-one function

Properties of Logarithms:

(1) $\log_a 1 = 0.$

(2) $\log_a(xy) = \log_a x + \log_a y.$

(3) $\log_a(\frac{x}{y}) = \log_a x - \log_a y.$

(4) $\log_a b^x = x \log_a b.$

(5) $a^{\log_a x} = x.$

(6) $\log_a b = \dfrac{\ln b}{\ln a} = \dfrac{\log_c b}{\log_c a}$ for $c > 0, \quad c \neq 1.$

Properties of Natural Logarithmic Function:

1. $\ln 1 = 0$.
2. $\ln(xy) = \ln x + \ln y$ for all positive real numbers x and y.
3. $\ln(\frac{x}{y}) = \ln x - \ln y$ for all positive real numbers x and y.
4. $\ln b^x = x \ln b$ for $b > 0$ and $x \in \mathbb{R}$.
5. $e^{\ln x} = x$ for each positive real number x.
6. $\ln(e^x) = x$ for every real number x.
7. $\log_a b = \dfrac{\ln b}{\ln a}$ for $a > 0$, $a \neq 1$ and $b > 0$.

Example 9. *Evaluate each logarithm at the indicated value of x. Use a calculator if you need it.*

1. $f(x) = \log_{10} x$ *at* $x = 10000$. ***Answer:*** $f(10000) = 4$.
2. $g(x) = \log_4 x$ *at* $x = \frac{1}{64}$. ***Answer:*** $f(\frac{1}{64}) = -3$.
3. $h(x) = \ln x$ *at* $x = 1$. ***Answer:*** $h(1) = 0$.
4. $k(x) = \ln x$ *at* $x = e$. ***Answer:*** $f(e) = 1$.
5. $f(x) = \log_{10} x$ *at* $x = 25$. ***Answer:*** $f(25) = 1.3979$.
6. $g(x) = \log_3 x$ *at* $x = 0.95$. ***Answer:*** $g(0.95) = -0.047$.
7. $h(x) = \ln x$ *at* $x = \frac{2}{3}$. ***Answer:*** $h(\frac{2}{3}) = -0.405$.
8. $k(x) = \log_7 x$ *at* $x = 65$. ***Answer:*** $f(65) = 2.145$.